

**Time:** 5 minutes**Outcome:** Compare private and social incentives and identify the efficient quantity.**Difficulty:** ●●○ Intermediate

Externalities

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THE CORE IDEA

An externality is a cost or benefit imposed on someone outside a transaction. Negative externalities make private decision makers ignore part of social cost; positive externalities make them ignore part of social benefit. Efficiency uses all marginal social costs and benefits.

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HOW TO RECOGNIZE IT

- Ask whether the spillover is a cost or benefit and whether it comes from production or consumption.
- Negative: $MSC > MPC$ or $MPB > MSB$, so the market quantity is too high.
- Positive: $MSB > MPB$ or $MPC > MSC$, so the market quantity is too low.
- Corrective taxes, subsidies, permits, regulation, or bargaining can internalize the spillover.

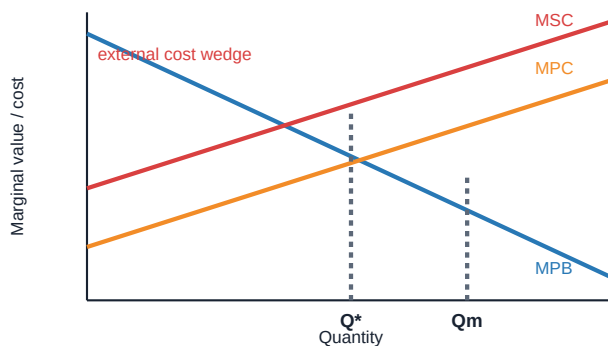
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WATCH OUT

Do not label every harmful outcome an externality. A spillover must affect a third party and be missing from the market price.

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WORKED EXAMPLE: NEGATIVE PRODUCTION EXTERNALITY



A factory's market supply is MPC. Pollution adds \$20 per unit, so MSC lies \$20 above MPC. The market chooses Q_m where $MPB = MPC$. The efficient quantity is the lower Q^* where $MPB = MSC$. A \$20 corrective tax aligns the firm's private marginal cost with social marginal cost.

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CHECK YOURSELF

If MSB lies above MPB, is the unregulated market quantity too high or too low?

**READY?**

Return to the game and master this concept.